

## Project Design Phase

### Problem – Solution Fit Template

|               |              |
|---------------|--------------|
| Date          | 26 June 2026 |
| Team ID       |              |
| Project Name  | Smart Lender |
| Maximum Marks | 2 Marks      |

#### Problem – Solution Fit Template:

Smart Lender is a Machine Learning-powered Loan Eligibility Prediction System that analyzes applicant information such as income, credit history, loan amount, education, employment status, and property area to predict loan approval eligibility.

#### Key Features:

- ☐ Automated loan eligibility prediction
- ☐ Real-time approval/rejection results
- ☐ XGBoost-based prediction engine
- ☐ Data preprocessing and feature engineering
- ☐ Interactive Flask web application
- ☐ Reduced manual effort and faster decision-making

#### Template:

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|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Define CS, fit into CC                   | <b>1. CUSTOMER SEGMENT(S)</b> <span>CS</span><br>Who is your customer?<br><b>Primary Customers:</b> <ul style="list-style-type: none"> <li>- Loan Applicants</li> <li>- Credit Officers</li> <li>- Financial Analysts</li> <li>- Banks and Financial Institutions</li> </ul> <b>Secondary Customers:</b> <ul style="list-style-type: none"> <li>- Loan Processing Teams</li> <li>- Risk Assessment Departments</li> </ul>                                                               | <b>6. CUSTOMER CONSTRAINTS</b> <span>CC</span><br><ul style="list-style-type: none"> <li>- Incomplete applicant information</li> <li>- Lack of credit history</li> <li>- Limited time for manual verification</li> <li>- Large number of applications</li> <li>- Regulatory compliance requirements</li> </ul>                                                                                                                                                               | <b>5. AVAILABLE SOLUTIONS</b> <span>AS</span><br>Current solutions: <ul style="list-style-type: none"> <li>- Manual loan verification</li> <li>- Traditional credit assessment</li> <li>- Spreadsheet-based analysis</li> <li>- Rule-based approval systems</li> </ul> Limitations: <ul style="list-style-type: none"> <li>- Time-consuming</li> <li>- Human errors</li> <li>- Inconsistent decisions</li> <li>- High operational cost</li> </ul> | Explore AS, differentiate         |
|                                          | <b>2. JOBS-TO-BE-DONE / PROBLEMS</b> <span>J&amp;P</span><br><ul style="list-style-type: none"> <li>- Loan Applicant: Determine loan eligibility quickly. Avoid lengthy approval processes. Receive transparent loan decisions.</li> <li>- Credit Officer: Evaluate applications efficiently. Reduce manual verification effort. Improve approval decision accuracy.</li> <li>- Financial Institution: Minimize loan defaults. Reduce NPAs. Increase operational efficiency.</li> </ul> | <b>9. PROBLEM ROOT CAUSE</b> <span>RC</span><br><ul style="list-style-type: none"> <li>- Manual loan evaluation is slow.</li> <li>- Large applicant volume increases workload.</li> <li>- Multiple eligibility factors require analysis.</li> <li>- Human decision-making may introduce errors.</li> <li>- Traditional methods cannot scale efficiently.</li> </ul>                                                                                                          | <b>7. BEHAVIOUR</b> <span>BE</span><br><ul style="list-style-type: none"> <li>- Loan Applicants: Submit loan applications online. Provide income and credit information. Check approval status frequently.</li> <li>- Credit Officers: Review applicant records. Verify financial documents. Assess risk before approval.</li> </ul>                                                                                                              |                                   |
| Focus on J&P, tap into BE, understand RC | <b>3. TRIGGERS</b> <span>TR</span><br><ul style="list-style-type: none"> <li>- Increase in loan applications.</li> <li>- Need for faster loan approval.</li> <li>- High volume of manual evaluations.</li> <li>- Rising loan default risks.</li> <li>- Demand for digital banking services.</li> </ul>                                                                                                                                                                                  | <b>10. YOUR SOLUTION</b> <span>SL</span><br><b>Smart Lender</b><br>Smart Lender is a Machine Learning-powered Loan Eligibility Prediction System that analyzes applicant information such as income, credit history, loan amount, education, employment status, and property area to predict loan approval eligibility.<br><b>Benefits:</b> Faster loan processing, Improved prediction accuracy, Reduced lending risk, Better customer experience, Lower operational costs. | <b>8. CHANNELS of BEHAVIOUR</b> <span>CH</span><br><b>8.1 ONLINE</b> <ul style="list-style-type: none"> <li>- Bank Website</li> <li>- Smart Lender Web Application</li> <li>- IBM Cloud Platform</li> <li>- Online Banking Portals</li> </ul> <b>8.2 OFFLINE</b> <ul style="list-style-type: none"> <li>- Bank Branches</li> <li>- Credit Officer Verification</li> <li>- Financial Document Submission</li> </ul>                                | Extract online & offline CH of BE |
|                                          | <b>4. EMOTIONS: BEFORE / AFTER</b> <span>EM</span><br><b>Before Using Smart Lender:</b> <ul style="list-style-type: none"> <li>- Uncertainty</li> <li>- Frustration</li> <li>- Trust in AI-based predictions</li> </ul>                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |
| Identify strong TR & EM                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>10. YOUR SOLUTION</b><br><b>Before Using Smart Lender:</b> <ul style="list-style-type: none"> <li>- Uncertainty</li> <li>- Anxiety</li> <li>- Frustration</li> <li>- Lack of confidence in manual processes</li> </ul> <b>After Using Smart Lender:</b> <ul style="list-style-type: none"> <li>- Confidence</li> <li>- Satisfaction</li> <li>- Faster decision-making</li> <li>- Trust in AI-based predictions</li> </ul>                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |